GRAVITATION

Class 9 | Science | Chapter 10

UNIVERSAL LAW OF GRAVITATION

Every object in the universe attracts every other object with a force proportional to the product of their masses and inversely proportional to the square of the distance between them.

$F = G \times (m_1 \times m_2) / d^2$

G = Universal gravitational constant = 6.674×10^{-11} N m^2/kg^2. Given by Sir Isaac Newton.

IMPORTANCE OF THE UNIVERSAL LAW

- Explains the force that binds us to the Earth.
- Explains the motion of the Moon around the Earth, and planets around the Sun.
- Explains tides due to the Moon and the Sun.

FREE FALL & ACCELERATION DUE TO GRAVITY

Formulae & Key Values

FREE FALL

When an object falls under the influence of gravity ALONE (no other force), it is said to be in free fall.

The acceleration produced is called acceleration due to gravity (g). It is independent of the mass of the falling object.

Quantity	Value
g (Earth, average)	$9.8\ m/s^2$ (approx. $10\ m/s^2$)
g on the Moon	About 1/6th of g on Earth
G (universal constant)	$6.674 \times 10^{-11}\ N\ m^2/kg^2$

EQUATIONS OF MOTION UNDER GRAVITY

$v = u + gt$

$h = ut + (1/2)gt^2$

$v^2 = u^2 + 2gh$

Use +g for a falling object (downward motion) and -g for an object thrown upward (motion against gravity).

MASS, WEIGHT & THRUST-PRESSURE

Key Definitions

MASS VS WEIGHT

Property	Mass	Weight
Meaning	Amount of matter in an object	Force of gravity acting on an object
Formula	-	W = m x g
Unit (SI)	kilogram (kg)	newton (N)
Nature	Scalar, constant everywhere	Vector, varies with g (location)
On the Moon	Same as on Earth	About 1/6th of weight on Earth

THRUST & PRESSURE

Thrust = the force acting on an object perpendicular to its surface.

Pressure = Thrust / Area, SI unit: pascal (Pa) = N/m^2

Same thrust over a smaller area produces greater pressure. Ex: sharp knife cuts easily; wide school-bag straps reduce pressure on shoulders.

ARCHIMEDES' PRINCIPLE

When an object is fully or partially immersed in a fluid, it experiences an upward buoyant force equal to the weight of the fluid displaced.

Objects with density LESS than the fluid float; objects with density MORE than the fluid sink.

SUMMARY & REVISION

Key Formulae + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Gravitation** -> universal force of attraction between all objects
- **Free fall** -> falling only under gravity
- **g** -> acceleration due to gravity (9.8 m/s^2 on Earth)
- **Weight** -> W = mg, force due to gravity
- **Thrust** -> perpendicular force on a surface
- **Pressure** -> thrust per unit area
- **Buoyancy** -> upward force by a fluid on an immersed object

SOLVED EXAMPLE

Q: Find the weight of a 5 kg object on Earth (g = 9.8 m/s^2).

A: W = m x g = 5 x 9.8 = 49 N.

ONE-LINE RECAP FOR REVISION

F = Gm1m2/d^2 -- gravity acts between every pair of objects.

Weight = mass x g; mass stays constant, weight changes with g.

Floating depends on comparing object density with fluid density.

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